

Advanced Functions

ANSWER KEY



Composite functions with restrictions

- 1 If $f(x) = \sqrt{x}$ and $g(x) = x - 5$, find the domain of $(f \circ g)(x)$.

$(f \circ g)(x) = \sqrt{x - 5}$, which requires $x - 5 \geq 0$, so the domain is $x \geq 5$.

- 2 If $f(x) = x^2 - 1$ and $g(x) = \frac{1}{x}$, find $(f \circ g)(x)$ and simplify.

$$(f \circ g)(x) = f\left(\frac{1}{x}\right) = \left(\frac{1}{x}\right)^2 - 1 = \frac{1}{x^2} - 1.$$

Inverse functions with domain restrictions

- 3 Find the inverse of $f(x) = x^2$ for $x \geq 0$, and explain why the restriction is necessary.

$f^{-1}(x) = \sqrt{x}$. The restriction $x \geq 0$ is necessary because $f(x) = x^2$ is not one-to-one over all real numbers (e.g. $f(2) = f(-2) = 4$), so it has no inverse function unless the domain is restricted to make it one-to-one.

End behavior of polynomial functions

- 4 Describe the end behavior of $f(x) = -2x^4 + 3x^2 - 1$, as $x \rightarrow \infty$ and $x \rightarrow -\infty$.

Since the leading term is $-2x^4$ (even degree, negative coefficient), both ends of the graph point downward: as $x \rightarrow \infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$.

- 5 Describe the end behavior of $g(x) = 3x^5 - x^3 + 2$.

Since the leading term is $3x^5$ (odd degree, positive coefficient): as $x \rightarrow \infty$, $g(x) \rightarrow \infty$; as $x \rightarrow -\infty$, $g(x) \rightarrow -\infty$.

Rational functions: asymptotes and holes

- 6 Find the vertical and horizontal asymptotes of $f(x) = \frac{2x + 1}{x - 3}$.

Vertical asymptote: where the denominator is zero: $x = 3$. Horizontal asymptote: since numerator and denominator have the same degree, the asymptote is the ratio of leading coefficients: $y = \frac{2}{1} = 2$.

- 7 Find the hole (removable discontinuity) in the graph of $f(x) = \frac{x^2 - 4}{x - 2}$.

Factor: $f(x) = \frac{(x - 2)(x + 2)}{x - 2} = x + 2$ for $x \neq 2$. There is a hole at $x = 2$, at the point $(2, 4)$ (found by substituting $x = 2$ into the simplified expression $x + 2$).

Logarithmic functions and their properties

- 8 Simplify $\log_3(9) + \log_3(27)$, using the product rule for logarithms.

$$\log_3(9) + \log_3(27) = \log_3(9 \times 27) = \log_3(243) = \log_3(3^5) = 5.$$

- 9 Solve $\ln(x) + \ln(x - 3) = \ln(10)$.

Combine logs: $\ln[x(x - 3)] = \ln(10)$, so $x(x - 3) = 10$, giving $x^2 - 3x - 10 = 0$. Factor: $(x - 5)(x + 2) = 0$, so $x = 5$ or $x = -2$. Since $\ln(x)$ requires $x > 0$, reject $x = -2$. Solution: $x = 5$.

Even, odd, and periodic function classification

- 10 Determine whether $f(x) = x^4 - 3x^2$ is even, odd, or neither.

$$f(-x) = (-x)^4 - 3(-x)^2 = x^4 - 3x^2 = f(x). \text{ Since } f(-x) = f(x), \text{ the function is even.}$$

Piecewise-defined functions and continuity

- 11 A function is defined as $f(x) = 3x + 1$ for $x < 2$ and $f(x) = kx - 3$ for $x \geq 2$. Find the value of k that makes f continuous at $x = 2$.

For continuity, the two pieces must agree at $x = 2$: $3(2) + 1 = k(2) - 3$, so $7 = 2k - 3$, giving $2k = 10$, $k = 5$.

A multi-part function transformation and composition problem

- 12 This question has two parts. Given $f(x) = 2x - 3$ and $g(x) = x^2$. (a) Find $(f \circ g)(x)$ and $(g \circ f)(x)$, in simplified form. (b) Find the value of x for which $(f \circ g)(x) = (g \circ f)(x)$.

(a) $(f \circ g)(x) = f(x^2) = 2x^2 - 3$. $(g \circ f)(x) = g(2x - 3) = (2x - 3)^2 = 4x^2 - 12x + 9$. (b) Set equal: $2x^2 - 3 = 4x^2 - 12x + 9$, so $0 = 2x^2 - 12x + 12$, giving $x^2 - 6x + 6 = 0$. Using the quadratic formula:

$$x = \frac{6 \pm \sqrt{36 - 24}}{2} = \frac{6 \pm \sqrt{12}}{2} = 3 \pm \sqrt{3}.$$

Finding zeros of higher-degree functions graphically and algebraically

- 13 A quartic function $f(x) = x^4 - 5x^2 + 4$ can be treated as a quadratic in x^2 . Find all real zeros.

Let $u = x^2$: $u^2 - 5u + 4 = 0$, factoring to $(u - 1)(u - 4) = 0$, so $u = 1$ or $u = 4$. Since $u = x^2$: $x^2 = 1$ gives $x = \pm 1$; $x^2 = 4$ gives $x = \pm 2$. All real zeros: $x = -2, -1, 1, 2$.

Function operations: sum, difference, product, quotient

- 14 Given $f(x) = x^2 + 1$ and $g(x) = x - 3$, find $(f + g)(x)$, $(f - g)(x)$, and $(fg)(x)$.

$$(f + g)(x) = x^2 + 1 + x - 3 = x^2 + x - 2. \quad (f - g)(x) = x^2 + 1 - (x - 3) = x^2 - x + 4.$$

$$(fg)(x) = (x^2 + 1)(x - 3) = x^3 - 3x^2 + x - 3.$$

Symmetry and transformations of rational functions

- 15 Explain why $f(x) = \frac{1}{x}$ is symmetric about the origin, and state whether it is even, odd, or neither.

$f(-x) = \frac{1}{-x} = -\frac{1}{x} = -f(x)$. Since $f(-x) = -f(x)$, the function is odd, meaning its graph has point symmetry about the origin — rotating the graph 180° about the origin maps it onto itself.

Finding a function's formula from a description of transformations

- 16 The graph of $y = \sqrt{x}$ is shifted 2 units right, stretched vertically by a factor of 3, and shifted 1 unit down. Write the equation of the resulting function.

Shifting right by 2 replaces x with $x - 2$: $\sqrt{x - 2}$. Vertical stretch by 3: $3\sqrt{x - 2}$. Shift down 1:
 $y = 3\sqrt{x - 2} - 1$.

A multi-part rational function analysis problem

- 17 This question has three parts. A function is given by $f(x) = \frac{2x^2 - 8}{x^2 - x - 6}$. (a) Factor the numerator and denominator completely. (b) Identify any holes and vertical asymptotes. (c) State the horizontal asymptote.

(a) Numerator: $2x^2 - 8 = 2(x^2 - 4) = 2(x - 2)(x + 2)$. Denominator: $x^2 - x - 6 = (x - 3)(x + 2)$. (b) The factor $(x + 2)$ cancels, indicating a hole at $x = -2$ (specifically at the point found by substituting into the simplified expression). The remaining denominator factor $(x - 3)$ gives a vertical asymptote at $x = 3$ (not canceled). (c) Since the numerator and denominator have the same degree (both degree 2), the horizontal asymptote is the ratio of leading coefficients: $y = \frac{2}{1} = 2$.

Finding a quadratic function from a table of values

- 18 A quadratic function $f(x) = ax^2 + bx + c$ has $f(0) = 5$, $f(1) = 10$, and $f(2) = 19$. Find a , b , and c .

From $f(0) = 5$: $c = 5$. From $f(1) = 10$: $a + b + c = 10$, so $a + b = 5$. From $f(2) = 19$: $4a + 2b + c = 19$, so $4a + 2b = 14$, i.e. $2a + b = 7$. Subtracting $a + b = 5$ from $2a + b = 7$: $a = 2$, then $b = 5 - 2 = 3$.
Function: $f(x) = 2x^2 + 3x + 5$.

Determining if a function has an inverse using the horizontal line test

- 19 Explain the horizontal line test, and use it to determine whether $f(x) = x^3$ has an inverse function.

The horizontal line test states that a function has an inverse (is one-to-one) if and only if no horizontal line intersects its graph more than once. For $f(x) = x^3$, every horizontal line $y = k$ intersects the graph exactly once (since every real number has exactly one real cube root), so $f(x) = x^3$ does have an inverse function, namely $f^{-1}(x) = \sqrt[3]{x}$.

Multiple choice — easy

- 20 Describe the end behavior of $f(x) = 4x^4 - x^3$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$

- a. $-\infty, \infty$
- b. $\infty, -\infty$
- c. $-\infty, -\infty$
- d. ∞, ∞ ✓

The leading term is $4x^4$ (even degree, positive coefficient), so as $x \rightarrow \infty$, $f(x) \rightarrow \infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow \infty$.

- 21 Describe the end behavior of $f(x) = -x^2 - 2x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$

- a. ∞, ∞
- b. $-\infty, \infty$
- c. $-\infty, -\infty$ ✓
- d. $\infty, -\infty$

The leading term is $-x^2$ (even degree, negative coefficient), so as $x \rightarrow \infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$.

22 Describe the end behavior of $f(x) = -x^3 + 8x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$

a. $\infty, -\infty$

b. $-\infty, \infty$ ✓

c. ∞, ∞

d. $-\infty, -\infty$

The leading term is $-x^3$ (odd degree, negative coefficient), so as $x \rightarrow \infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow \infty$.

23 Describe the end behavior of $f(x) = -2x^2 + 9x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$

a. ∞, ∞

b. $\infty, -\infty$

c. $-\infty, -\infty$ ✓

d. $-\infty, \infty$

The leading term is $-2x^2$ (even degree, negative coefficient), so as $x \rightarrow \infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$.

24 Describe the end behavior of $f(x) = 4x^2 + 4x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$

a. $\infty, -\infty$

b. $-\infty, -\infty$

c. $-\infty, \infty$

d. ∞, ∞ ✓

The leading term is $4x^2$ (even degree, positive coefficient), so as $x \rightarrow \infty$, $f(x) \rightarrow \infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow \infty$.

25 Describe the end behavior of $f(x) = -4x^4 - 2x^2$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$

- a. $-\infty, \infty$
- b. $\infty, -\infty$
- c. ∞, ∞
- d. $-\infty, -\infty$ ✓

The leading term is $-4x^4$ (even degree, negative coefficient), so as $x \rightarrow \infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$.

26 Describe the end behavior of $f(x) = 4x^5 + 5x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$

- a. $-\infty, -\infty$
- b. $\infty, -\infty$ ✓
- c. ∞, ∞
- d. $-\infty, \infty$

The leading term is $4x^5$ (odd degree, positive coefficient), so as $x \rightarrow \infty$, $f(x) \rightarrow \infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$.

27 Describe the end behavior of $f(x) = x^2 - 2x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$

- a. ∞, ∞ ✓
- b. $-\infty, -\infty$
- c. $-\infty, \infty$
- d. $\infty, -\infty$

The leading term is x^2 (even degree, positive coefficient), so as $x \rightarrow \infty$, $f(x) \rightarrow \infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow \infty$.

28 Describe the end behavior of $f(x) = x^2 + 9x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$

a. $\infty, -\infty$

b. $-\infty, -\infty$

c. $-\infty, \infty$

d. ∞, ∞ ✓

The leading term is x^2 (even degree, positive coefficient), so as $x \rightarrow \infty$, $f(x) \rightarrow \infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow \infty$.

29 Describe the end behavior of $f(x) = x^4 + 9x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$

a. $-\infty, \infty$

b. ∞, ∞ ✓

c. $\infty, -\infty$

d. $-\infty, -\infty$

The leading term is x^4 (even degree, positive coefficient), so as $x \rightarrow \infty$, $f(x) \rightarrow \infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow \infty$.

30 Describe the end behavior of $f(x) = -4x^2 + 4x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$

a. ∞, ∞

b. $-\infty, -\infty$ ✓

c. $\infty, -\infty$

d. $-\infty, \infty$

The leading term is $-4x^2$ (even degree, negative coefficient), so as $x \rightarrow \infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$.

31 Describe the end behavior of $f(x) = x^5 + 5x^3$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$

- a. $-\infty, \infty$
- b. $\infty, -\infty$ ✓
- c. $-\infty, -\infty$
- d. ∞, ∞

The leading term is x^5 (odd degree, positive coefficient), so as $x \rightarrow \infty$, $f(x) \rightarrow \infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$.

32 Describe the end behavior of $f(x) = x^5 - x^3$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$

- a. $\infty, -\infty$ ✓
- b. $-\infty, \infty$
- c. $-\infty, -\infty$
- d. ∞, ∞

The leading term is x^5 (odd degree, positive coefficient), so as $x \rightarrow \infty$, $f(x) \rightarrow \infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$.

33 Describe the end behavior of $f(x) = -3x^2 - 9x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$

- a. $-\infty, \infty$
- b. $\infty, -\infty$
- c. $-\infty, -\infty$ ✓
- d. ∞, ∞

The leading term is $-3x^2$ (even degree, negative coefficient), so as $x \rightarrow \infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$.

34 Describe the end behavior of $f(x) = -4x^2 + 7x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$

a. $-\infty, -\infty$ ✓

b. $-\infty, \infty$

c. ∞, ∞

d. $\infty, -\infty$

The leading term is $-4x^2$ (even degree, negative coefficient), so as $x \rightarrow \infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$.

Multiple choice — medium

35 Find the vertical asymptote of $f(x) = \frac{x-15}{x+14}$.

a. -13

b. -15

c. 14

d. -14 ✓

The vertical asymptote occurs where the denominator is zero: $x + 14 = 0$, so $x = -14$.

36 Find the vertical asymptote of $f(x) = \frac{2x-13}{-x+10}$.

a. 9

b. 11

c. 10 ✓

d. -10

The vertical asymptote occurs where the denominator is zero: $-x + 10 = 0$, so $x = 10$.

37 Find the vertical asymptote of $f(x) = \frac{x-7}{2x-12}$.

- a. 7
- b. -6
- c. 6 ✓
- d. 5

The vertical asymptote occurs where the denominator is zero: $2x - 12 = 0$, so $x = 6$.

38 Find the vertical asymptote of $f(x) = \frac{2x+11}{-x-8}$.

- a. -7
- b. 8
- c. -9
- d. -8 ✓

The vertical asymptote occurs where the denominator is zero: $-x - 8 = 0$, so $x = -8$.

39 Find the vertical asymptote of $f(x) = \frac{7x+1}{3x+6}$.

- a. 2
- b. -3
- c. -2 ✓
- d. -1

The vertical asymptote occurs where the denominator is zero: $3x + 6 = 0$, so $x = -2$.

40 Find the vertical asymptote of $f(x) = \frac{-8x-1}{7x+7}$.

- a. 1
- b. -1 ✓
- c. 0
- d. -2

The vertical asymptote occurs where the denominator is zero: $7x + 7 = 0$, so $x = -1$.

41 Find the vertical asymptote of $f(x) = \frac{8x - 7}{-5x - 5}$.

- a. -2
- b. 0
- c. -1 ✓
- d. 1

The vertical asymptote occurs where the denominator is zero: $-5x - 5 = 0$, so $x = -1$.

42 Find the vertical asymptote of $f(x) = \frac{-8x + 1}{x}$.

- a. -1
- b. 0 ✓
- c. 1
- d. 2

The vertical asymptote occurs where the denominator is zero: $x = 0$, so $x = 0$.

43 Find the vertical asymptote of $f(x) = \frac{9x - 10}{x - 7}$.

- a. -7
- b. 7 ✓
- c. 6
- d. 8

The vertical asymptote occurs where the denominator is zero: $x - 7 = 0$, so $x = 7$.

44 Find the vertical asymptote of $f(x) = \frac{-2x + 6}{3x}$.

a. 0 ✓

b. -2

c. -1

d. 1

The vertical asymptote occurs where the denominator is zero: $3x = 0$, so $x = 0$.

45 Find the vertical asymptote of $f(x) = \frac{5x + 4}{-5x - 15}$.

a. -4

b. 3

c. -3 ✓

d. -2

The vertical asymptote occurs where the denominator is zero: $-5x - 15 = 0$, so $x = -3$.

46 Find the vertical asymptote of $f(x) = \frac{-9x - 12}{-x - 6}$.

a. -7

b. -5

c. 6

d. -6 ✓

The vertical asymptote occurs where the denominator is zero: $-x - 6 = 0$, so $x = -6$.

47 Find the vertical asymptote of $f(x) = \frac{-7x + 14}{-2x + 14}$.

a. 8

b. 7 ✓

c. -7

d. 6

The vertical asymptote occurs where the denominator is zero: $-2x + 14 = 0$, so $x = 7$.

48 Find the vertical asymptote of $f(x) = \frac{x - 12}{7x - 14}$.

a. -2

b. 2 ✓

c. 3

d. 1

The vertical asymptote occurs where the denominator is zero: $7x - 14 = 0$, so $x = 2$.

49 Find the vertical asymptote of $f(x) = \frac{7x + 14}{3x - 9}$.

a. 2

b. 4

c. -3

d. 3 ✓

The vertical asymptote occurs where the denominator is zero: $3x - 9 = 0$, so $x = 3$.

Multiple choice — hard

50 Solve $\ln(x) + \ln(x - 5) = \ln(55)$.

a. 11 ✓

b. 5

c. 10

d. 12

Combine logs: $\ln[x(x - 5)] = \ln(55)$, so $x(x - 5) = 55$, giving $x^2 - 5x - 55 = 0$. Factor:
 $(x - 11)(x - (-5)) = 0$, so $x = 11$ or $x = -5$. Since $\ln(x)$ requires $x > 0$, reject $x = -5$. Solution: $x = 11$.

51 Solve $\ln(x) + \ln(x - 1) = \ln(5)$.

a. 4

b. 1

c. 6

d. 5 ✓

Combine logs: $\ln[x(x - 1)] = \ln(5)$, so $x(x - 1) = 5$, giving $x^2 - x - 5 = 0$. Factor:
 $(x - 5)(x - (-1)) = 0$, so $x = 5$ or $x = -1$. Since $\ln(x)$ requires $x > 0$, reject $x = -1$. Solution: $x = 5$.

52 Solve $\ln(x) + \ln(x - 9) = \ln(72)$.

a. 9

b. 6

c. 8 ✓

d. 7

Combine logs: $\ln[x(x - 9)] = \ln(72)$, so $x(x - 9) = 72$, giving $x^2 - 9x - 72 = 0$. Factor:
 $(x - 8)(x - (-9)) = 0$, so $x = 8$ or $x = -9$. Since $\ln(x)$ requires $x > 0$, reject $x = -9$. Solution: $x = 8$.

53 Solve $\ln(x) + \ln(x - 8) = \ln(32)$.

a. 3

b. 8

c. 4 ✓

d. 5

Combine logs: $\ln[x(x - 8)] = \ln(32)$, so $x(x - 8) = 32$, giving $x^2 - 8x - 32 = 0$. Factor:
 $(x - 4)(x - (-8)) = 0$, so $x = 4$ or $x = -8$. Since $\ln(x)$ requires $x > 0$, reject $x = -8$. Solution: $x = 4$.

54 Solve $\ln(x) + \ln(x - 1) = \ln(7)$.

- a. 8
- b. 6
- c. 1
- d. 7 ✓

Combine logs: $\ln[x(x - 1)] = \ln(7)$, so $x(x - 1) = 7$, giving $x^2 - x - 7 = 0$. Factor:
 $(x - 7)(x - (-1)) = 0$, so $x = 7$ or $x = -1$. Since $\ln(x)$ requires $x > 0$, reject $x = -1$. Solution: $x = 7$.

55 The quartic function $f(x) = x^4 - 5x^2 + 4$ can be treated as a quadratic in x^2 . Find all real zeros.

- a. $-2, -1, 1, 2$ ✓
- b. $-1, -2, 1, 2$
- c. $1, 2$
- d. $-2, 2$

Let $u = x^2$: $u^2 - 5u + 4 = 0$, factoring to $(u - 1)(u - 4) = 0$, so $u = 1$ or $u = 4$. Since $u = x^2$: $x^2 = 1$ gives $x = \pm 1$; $x^2 = 4$ gives $x = \pm 2$. All real zeros: $x = -2, -1, 1, 2$.

56 The quartic function $f(x) = x^4 - 25x^2 + 144$ can be treated as a quadratic in x^2 . Find all real zeros.

- a. $3, 4$
- b. $-4, 4$
- c. $-3, -4, 3, 4$
- d. $-4, -3, 3, 4$ ✓

Let $u = x^2$: $u^2 - 25u + 144 = 0$, factoring to $(u - 9)(u - 16) = 0$, so $u = 9$ or $u = 16$. Since $u = x^2$: $x^2 = 9$ gives $x = \pm 3$; $x^2 = 16$ gives $x = \pm 4$. All real zeros: $x = -4, -3, 3, 4$.

57 The quartic function $f(x) = x^4 - 13x^2 + 36$ can be treated as a quadratic in x^2 . Find all real zeros.

- a. 2, 3
- b. -2, -3, 2, 3
- c. -3, -2, 2, 3 ✓
- d. -3, 3

Let $u = x^2$: $u^2 - 13u + 36 = 0$, factoring to $(u - 4)(u - 9) = 0$, so $u = 4$ or $u = 9$. Since $u = x^2$: $x^2 = 4$ gives $x = \pm 2$; $x^2 = 9$ gives $x = \pm 3$. All real zeros: $x = -3, -2, 2, 3$.

58 The quartic function $f(x) = x^4 - 17x^2 + 16$ can be treated as a quadratic in x^2 . Find all real zeros.

- a. -1, -4, 1, 4
- b. 1, 4
- c. -4, -1, 1, 4 ✓
- d. -4, 4

Let $u = x^2$: $u^2 - 17u + 16 = 0$, factoring to $(u - 1)(u - 16) = 0$, so $u = 1$ or $u = 16$. Since $u = x^2$: $x^2 = 1$ gives $x = \pm 1$; $x^2 = 16$ gives $x = \pm 4$. All real zeros: $x = -4, -1, 1, 4$.

59 The quartic function $f(x) = x^4 - 20x^2 + 64$ can be treated as a quadratic in x^2 . Find all real zeros.

- a. -4, 4
- b. -2, -4, 2, 4
- c. -4, -2, 2, 4 ✓
- d. 2, 4

Let $u = x^2$: $u^2 - 20u + 64 = 0$, factoring to $(u - 4)(u - 16) = 0$, so $u = 4$ or $u = 16$. Since $u = x^2$: $x^2 = 4$ gives $x = \pm 2$; $x^2 = 16$ gives $x = \pm 4$. All real zeros: $x = -4, -2, 2, 4$.

60 If $f(x) = -3x + 5$ and $g(x) = x^2 + 5$, find $(f \circ g)(3)$.

a. -39

b. -37 ✓

c. -35

d. 14

First, $g(3) = (3)^2 + 5 = 14$. Then $f(14) = -3(14) + 5 = -37$.

61 If $f(x) = 3x + 6$ and $g(x) = x^2 + 4$, find $(f \circ g)(5)$.

a. 29

b. 93 ✓

c. 91

d. 95

First, $g(5) = (5)^2 + 4 = 29$. Then $f(29) = 3(29) + 6 = 93$.

62 If $f(x) = -3x - 3$ and $g(x) = x^2 - 3$, find $(f \circ g)(1)$.

a. 5

b. 1

c. 3 ✓

d. -2

First, $g(1) = (1)^2 - 3 = -2$. Then $f(-2) = -3(-2) - 3 = 3$.

63 If $f(x) = 2x - 1$ and $g(x) = -2x^2$, find $(f \circ g)(-1)$.

a. -7

b. -2

c. -3

d. -5 ✓

First, $g(-1) = -2(-1)^2 = -2$. Then $f(-2) = 2(-2) - 1 = -5$.

64 If $f(x) = -2x + 4$ and $g(x) = 3x^2 - 4$, find $(f \circ g)(-5)$.

a. 71

b. -136

c. -138 ✓

d. -140

First, $g(-5) = 3(-5)^2 - 4 = 71$. Then $f(71) = -2(71) + 4 = -138$.